



Assignment

Subject: Introduction to Cyber Security

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Scenario # 1 Hospital Web Portal Attack

1. Role of Network / Internet in Enabling the Attack

The Internet played a major role in delivering and executing the attack:

- The hospital web portal was connected to the public Internet, allowing external access.
- The attacker used email communication over the Internet to send a spear-phishing message.
- The malicious JavaScript file was downloaded through the network when opened.
- The malware connected to a Command and Control (C2) server through the Internet.
- Patient records were exfiltrated remotely using network communication channels.

Therefore, network connectivity enabled attack delivery, remote control, and data theft.

2. CIA Triad Violations

CIA Component	Violation
Confidentiality	Patient records were stolen and exposed.
Integrity	Access logs were deleted and system records manipulated.
Availability	System reliability reduced due to malware activity.

3. Malware Attack Lifecycle (7 Stages)

1. Reconnaissance

Attacker identifies hospital using outdated Apache server.

2. Weaponization

Malicious JavaScript embedded inside fake staff roster file.

3. Delivery

Spear-phishing email sent to nurse.

4. Exploitation

File execution triggers drive-by download vulnerability.

5. Installation

Trojan malware installs on work laptop.

6. Command & Control

Malware connects to attacker's remote server.

7. Actions on Objectives

- Data exfiltration
- Log deletion
- Unauthorized access to records

4. Malware Type and Payload Classification

Malware Type

- Trojan Horse
- Social-engineering based malware

Payload Types

- Data exfiltration payload
- Privilege escalation payload
- Remote access payload
- Log destruction payload

5. Technical Defenses (4)

- Email filtering and phishing detection systems
- Regular server patching and updates
- Endpoint antivirus & EDR solutions
- Encryption of patient records and secure logging

Scenario # 2 — University Exam System Attack

1. Role of Network / Internet in Attack Delivery

The Internet enabled the attack in multiple ways:

- Attacker sent bulk phishing email through Internet mail servers.
- Fake link hosted on remote website downloaded malware.
- Keylogger transmitted captured credentials via network.
- Attacker remotely accessed university database using stolen login credentials.

Thus, the network allowed credential theft and remote database compromise.

2. CIA Triad Violations

CIA Component	Violation
Confidentiality	Exam answer sheets were exported illegally.
Integrity	Final grades of 500 students were altered.
Availability	Trust and reliability of exam system affected.

3. Malware Attack Lifecycle (7 Stages)

1. Reconnaissance

Attacker studies university system and professors.

2. Weaponization

Keylogger malware prepared.

3. Delivery

Phishing email with fake deadline extension link.

4. Exploitation

Professor downloads and runs malware.

5. Installation

Keylogger installs silently.

6. Command & Control

Captured credentials sent to attacker server.

7. Actions on Objectives

- Admin login access
- Grade manipulation
- Data export

4. Malware Type and Payload Classification

Malware Type

- Keylogger (Spyware category)

Payload Types

- Credential harvesting payload
- Data theft payload
- Unauthorized database access payload
- Academic record manipulation payload

5. Technical Countermeasures (4)

- Multi-Factor Authentication (MFA) for admin accounts
- Secure email gateway with phishing protection
- Regular software updates (PHP & MySQL patching)
- Database activity monitoring and logging systems